

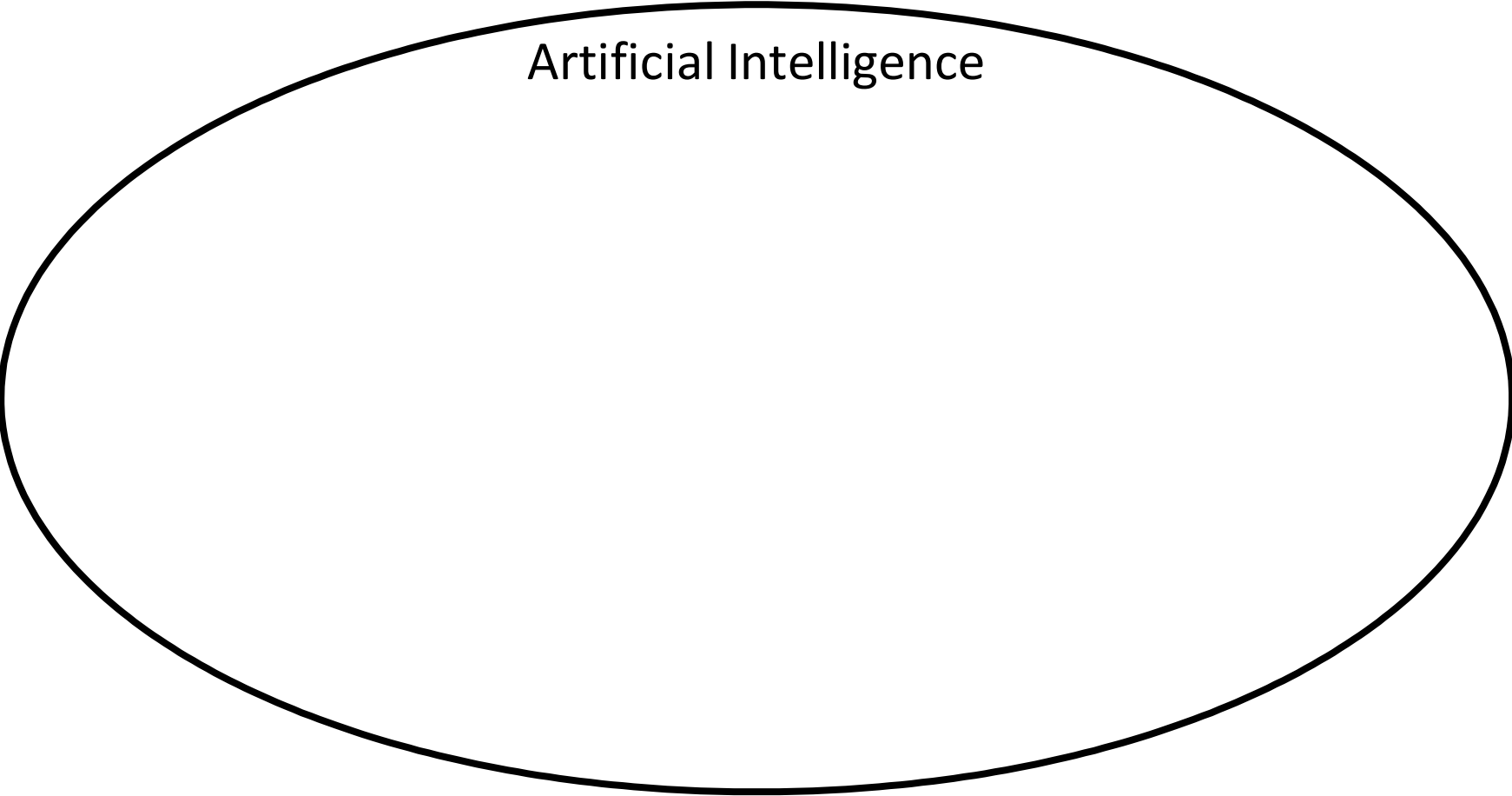
Introduction to Computer Vision

Course Code: CSC 4254 Course Title: Computer Vision & Pattern Recognition

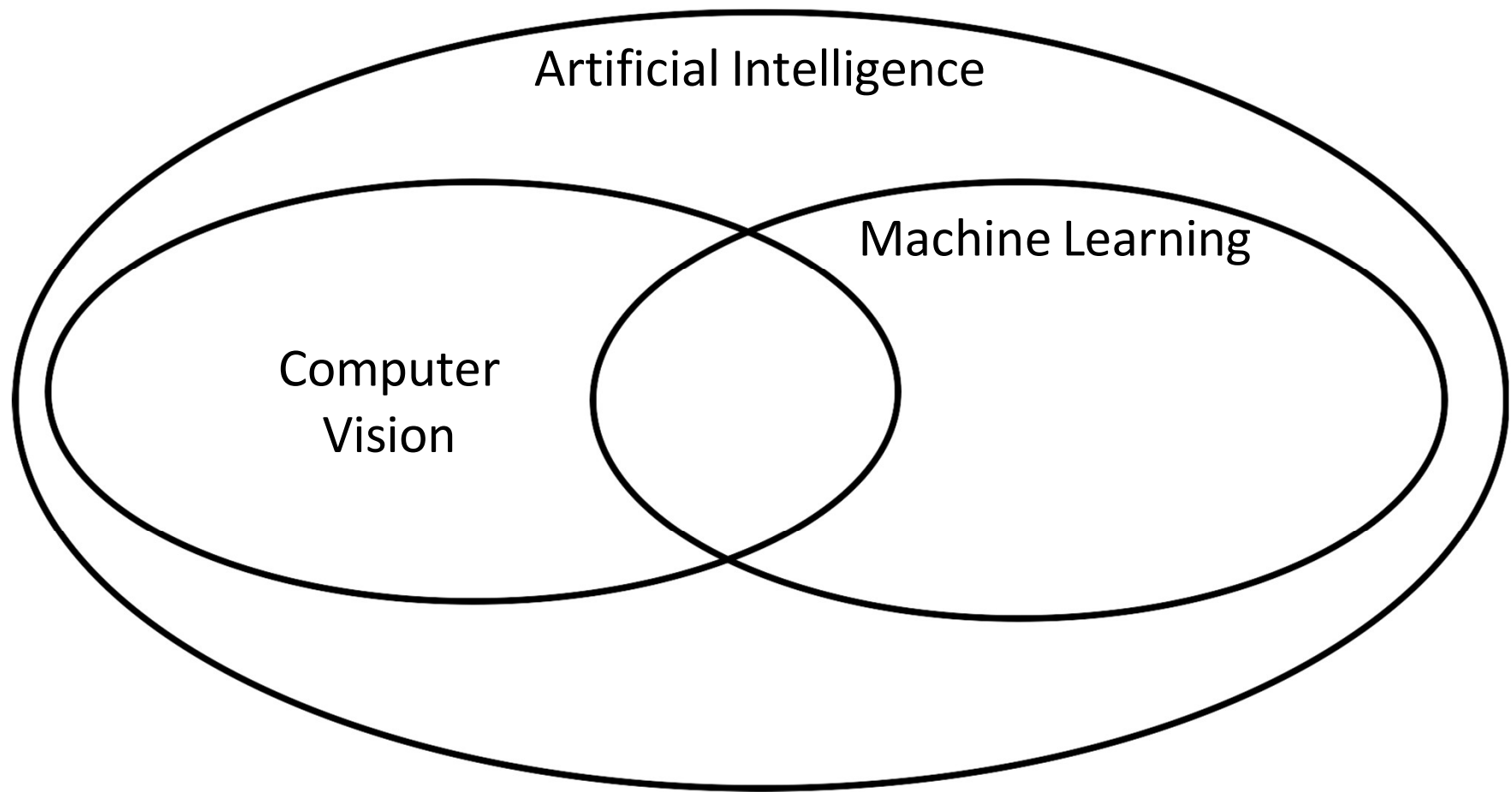


Dept. of Computer Science
Faculty of Science and Technology

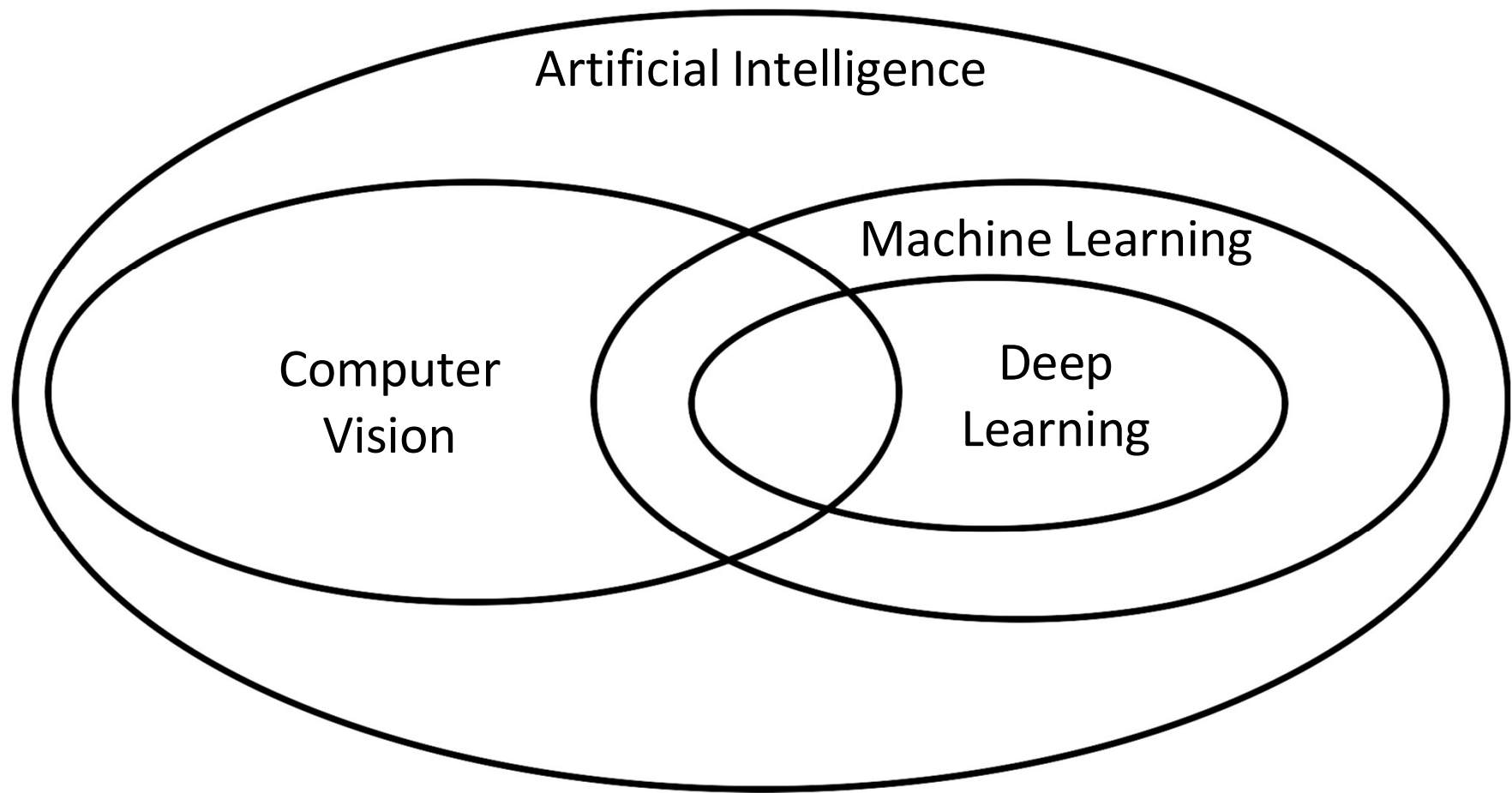
Lecturer No:	1	Week No:	1	Semester:	
Lecturer:					



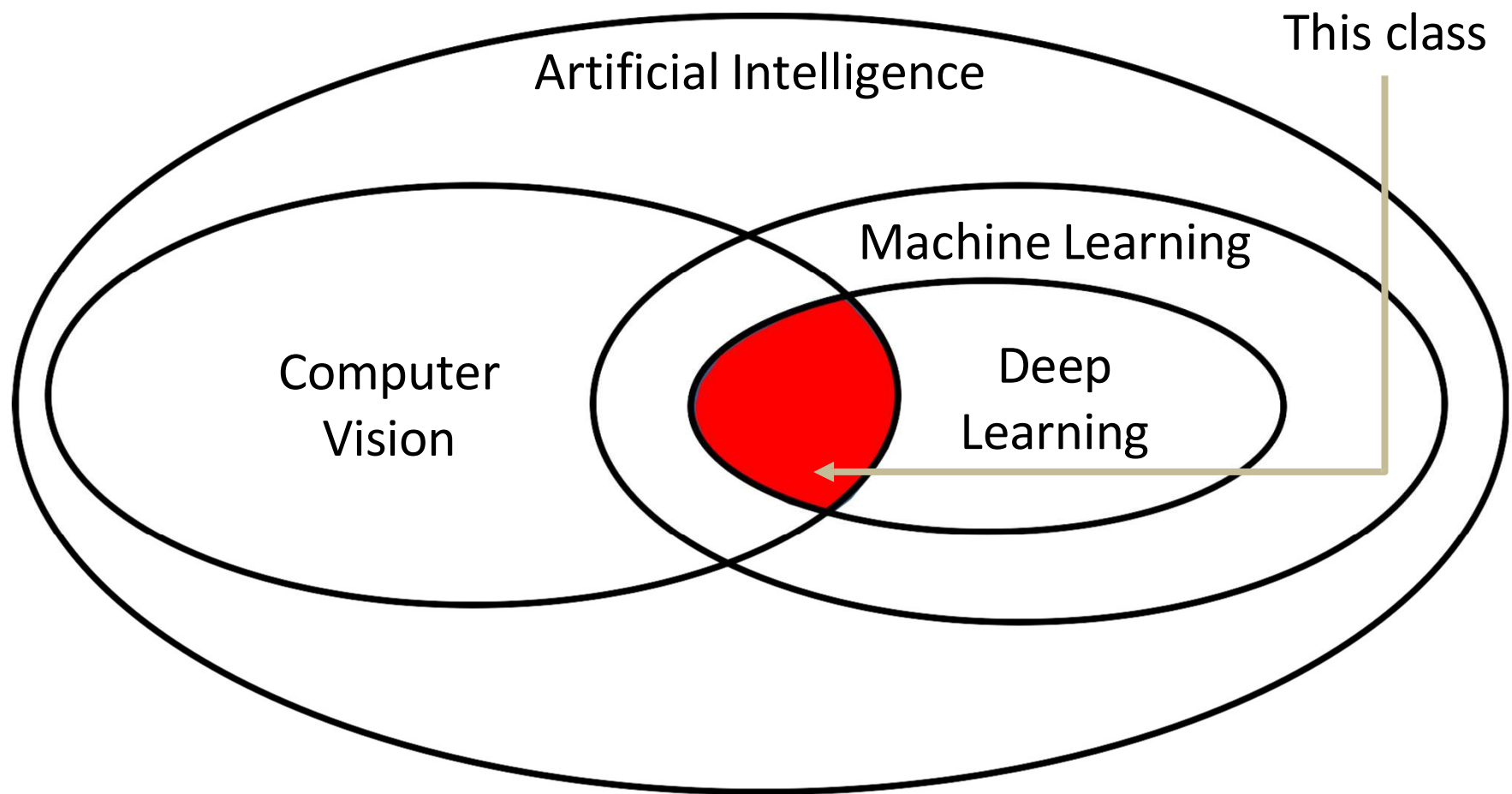
Artificial Intelligence



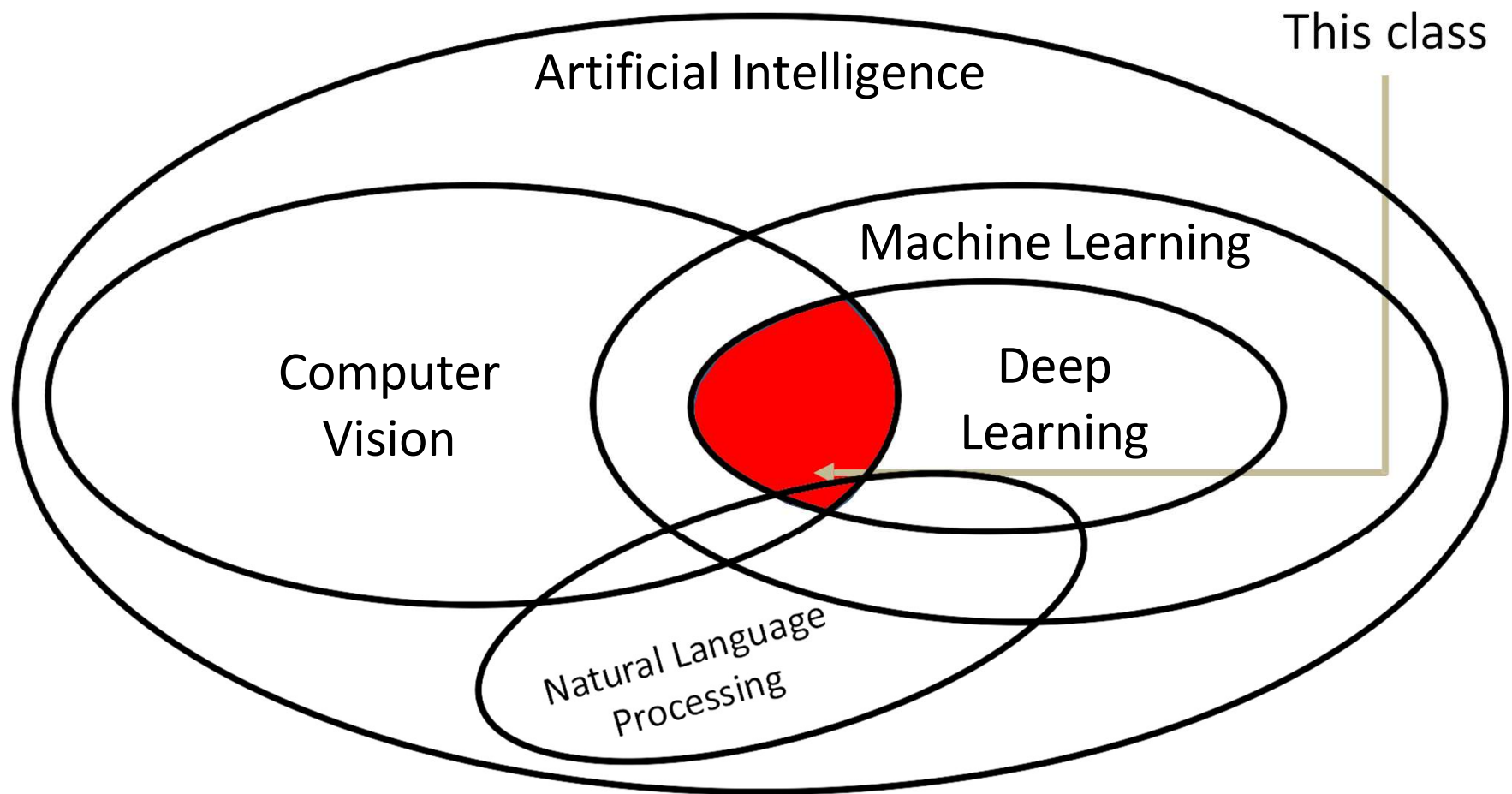
Slide inspiration: Justin Johnson



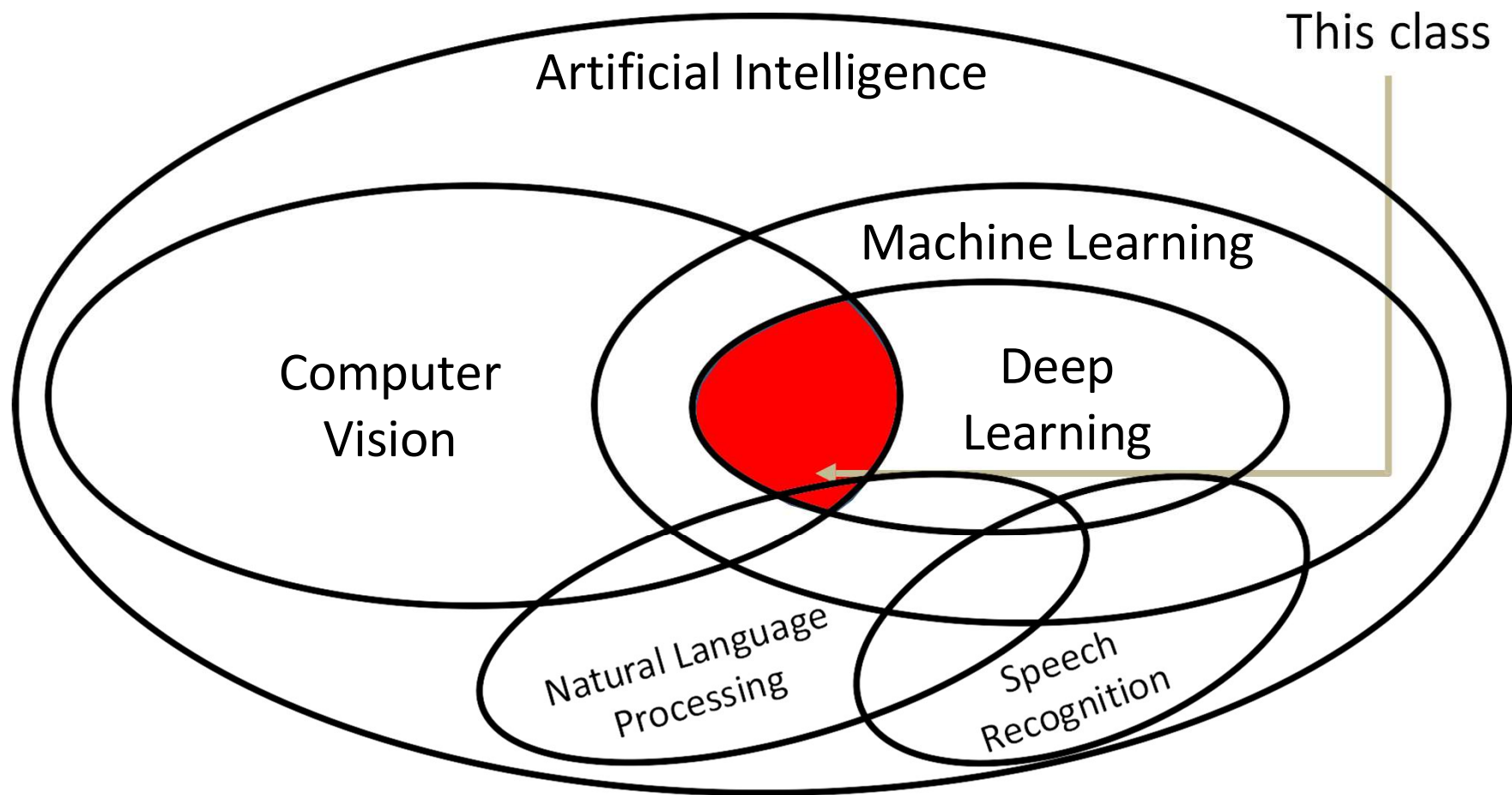
Slide inspiration: Justin Johnson



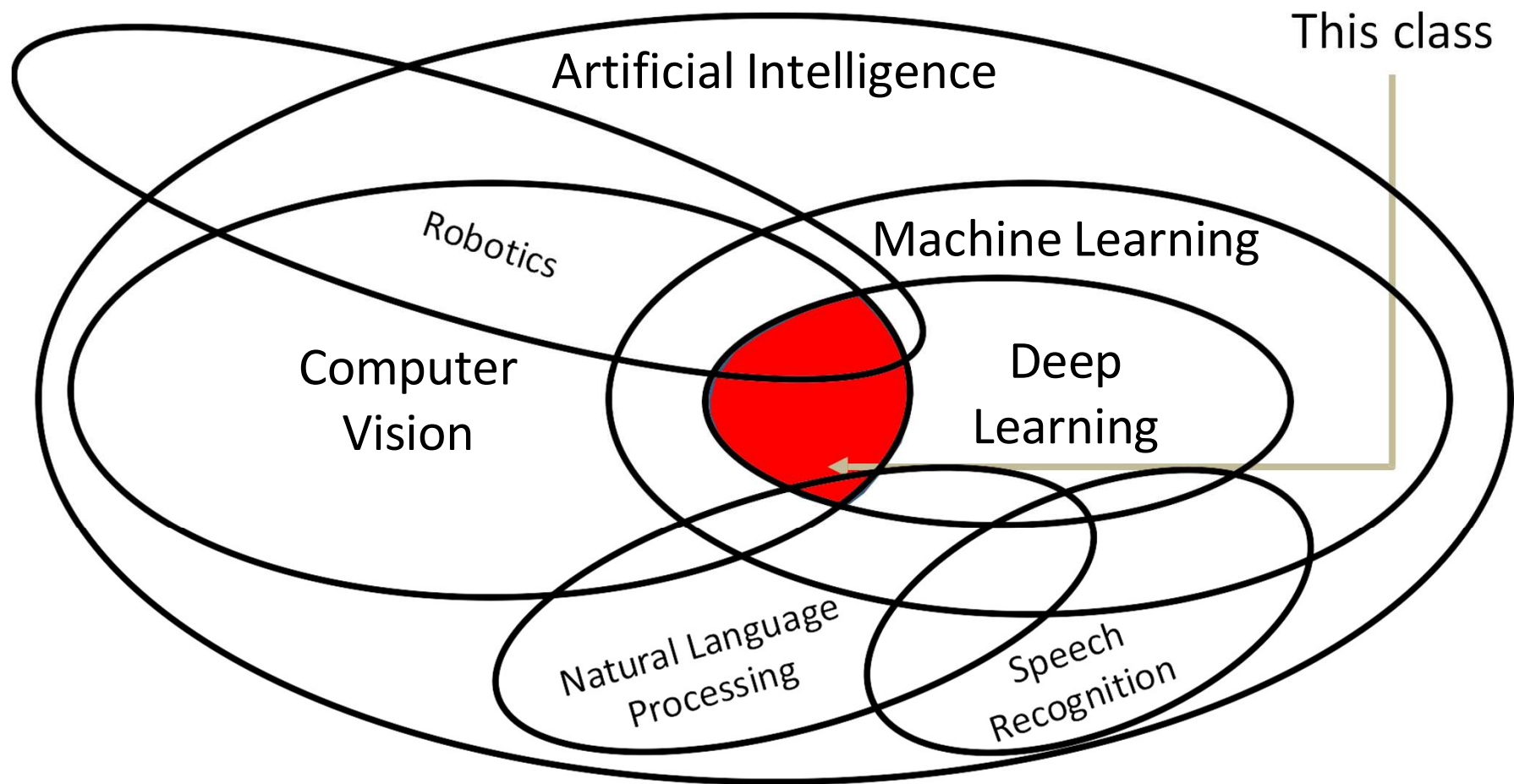
Slide inspiration: Justin Johnson



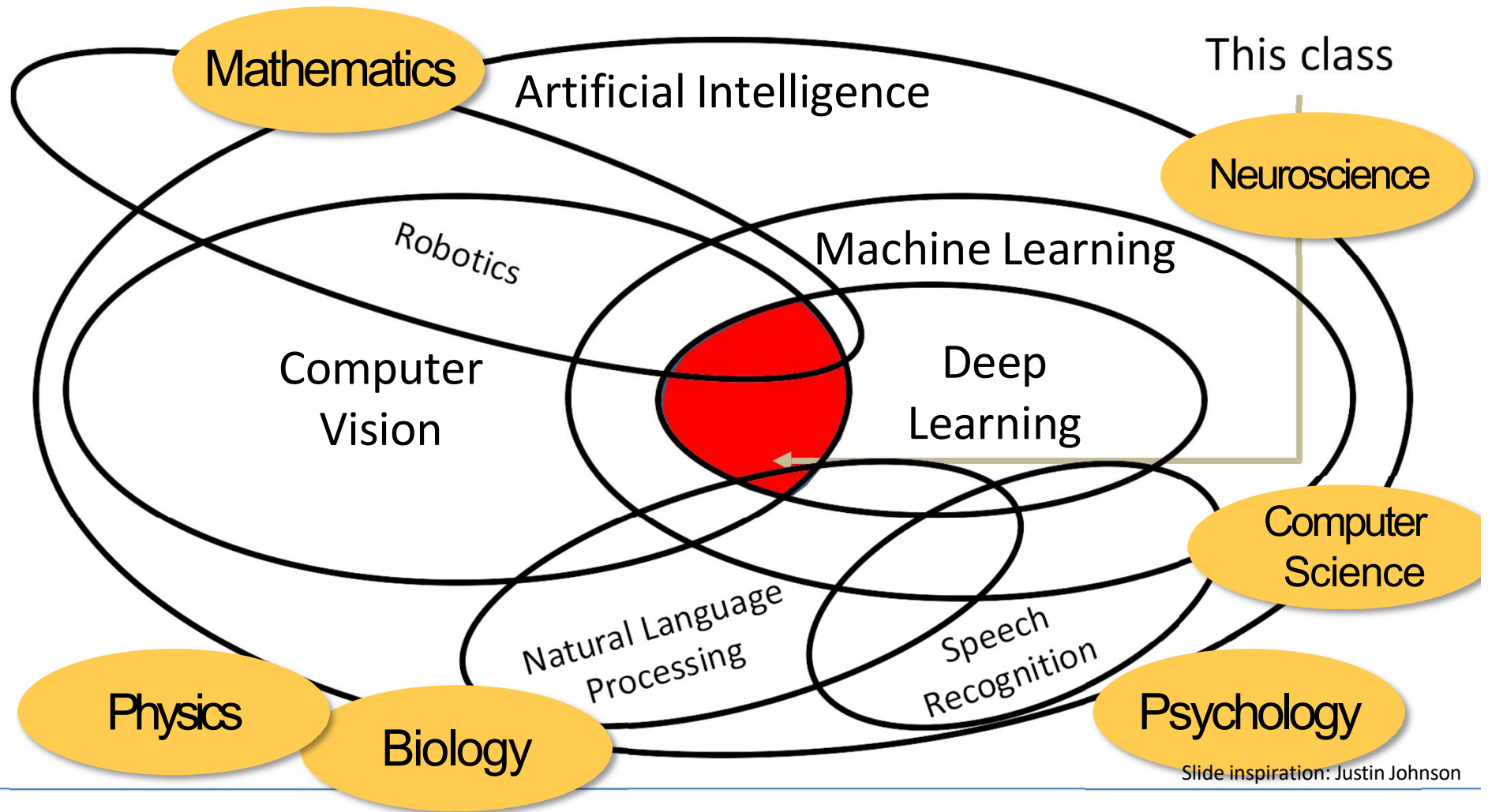
Slide inspiration: Justin Johnson



Slide inspiration: Justin Johnson



Slide inspiration: Justin Johnson



Slide inspiration: Justin Johnson

Today's agenda

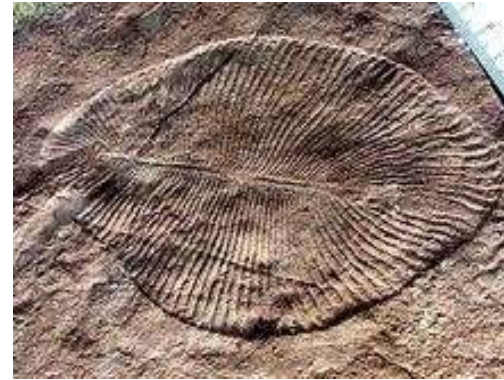
- A brief history of computer vision and deep learning
- Course overview



Evolution's Big Bang: Cambrian Explosion, 530-540million years, B.C.



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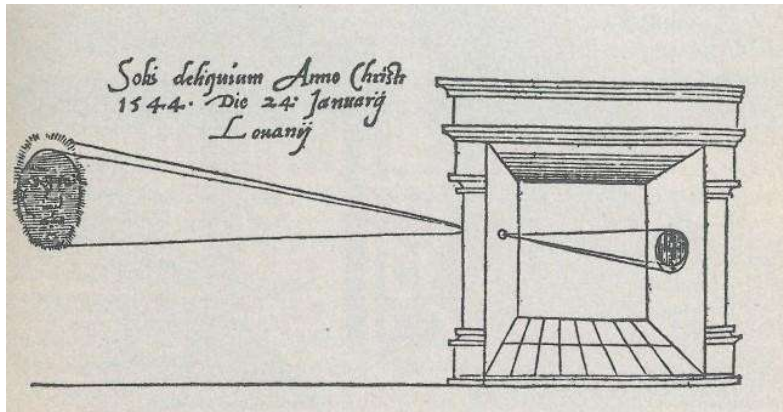


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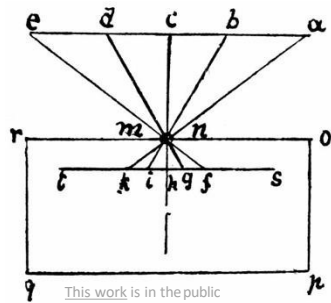


Camera Obscura

Gemma Frisius, 1545



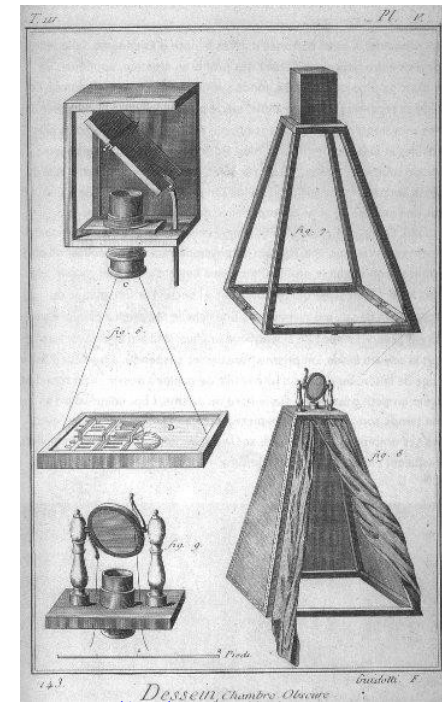
[This work](#) is in the public domain



[This work](#) is in the public domain

Leonardo da Vinci,
16th Century AD

Encyclopedia, 18th Century



[This work](#) is in the public domain

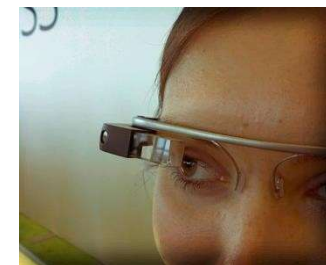
Computer Vision is everywhere!



Left to right:
 Image by Roger H Goun is licensed under CC BY 2.0
 Image is CC0 1.0 public domain
 Image is CC0 1.0 public domain
 Image is CC0 1.0 public domain

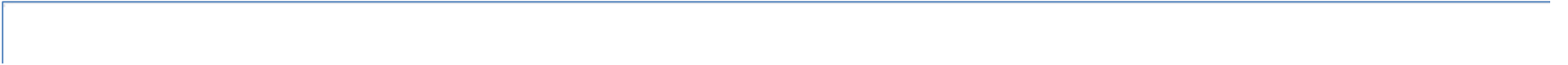


Left to right:
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Bottom row, left to right
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Where did we come from?



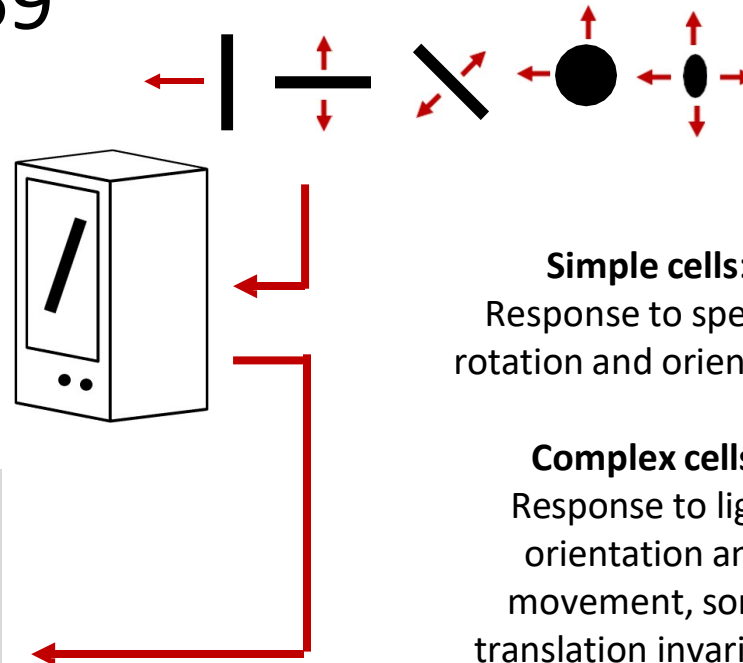
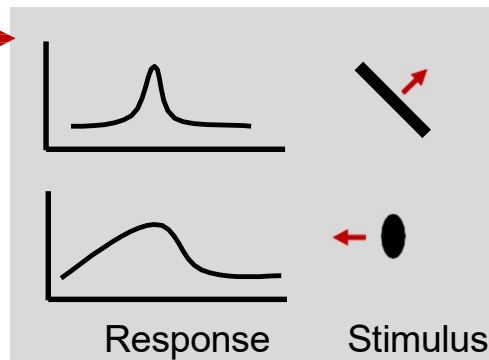
Hubel and Wiesel, 1959

Measure
brain activity



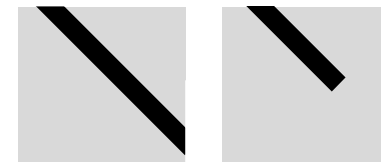
Cat image by CNX OpenStax is licensed under CC BY 4.0. changes made

1959
Hubel & Wiesel



Simple cells:
Response to specific
rotation and orientation

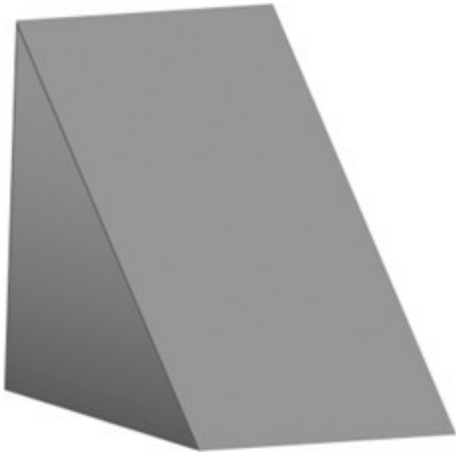
Complex cells:
Response to light
orientation and
movement, some
translation invariance



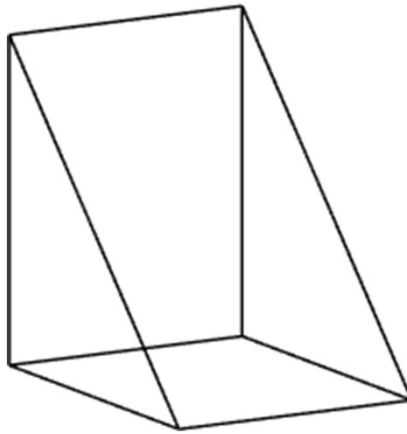
No response

Slide inspiration: Justin Johnson

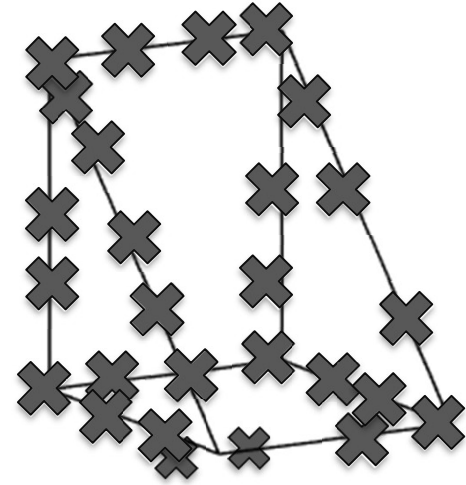
Larry Roberts, 1963



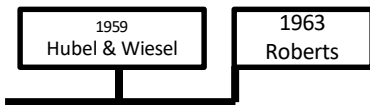
(a) Original picture



(b) Differentiated picture



(c) Feature points selected



MASSACHUSETTS INSTITUTE OF TECHNOLOGY
PROJECT MAC

Artificial Intelligence Group
Vision Memo. No. 100.

July 7, 1966

THE SUMMER VISION PROJECT

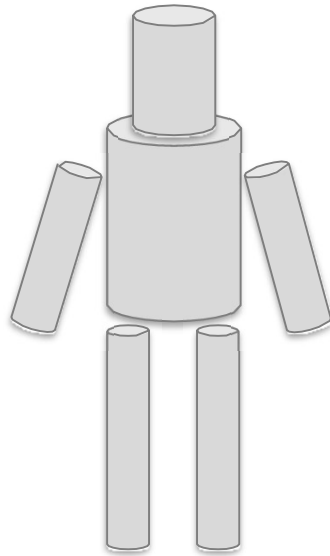
Seymour Papert

The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".

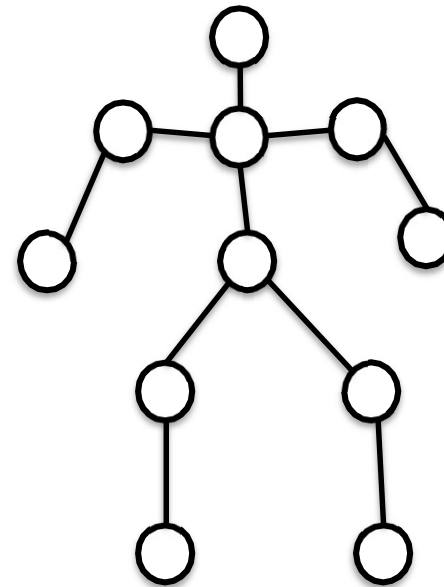
1959
Hubel & Wiesel

1963
Roberts

Recognition via Parts (1970s)



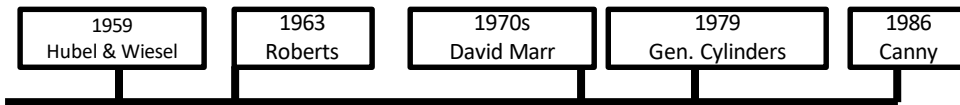
Generalized Cylinders,
Brooks and Binford,
1979



Pictorial Structures,
Fischler and Elshlager, 1973

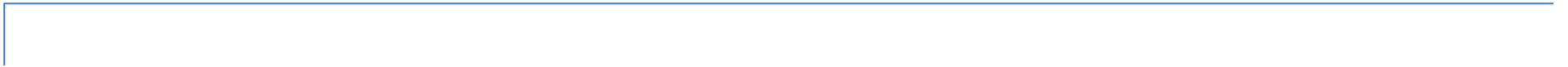


Recognition via Edge Detection (1980s)



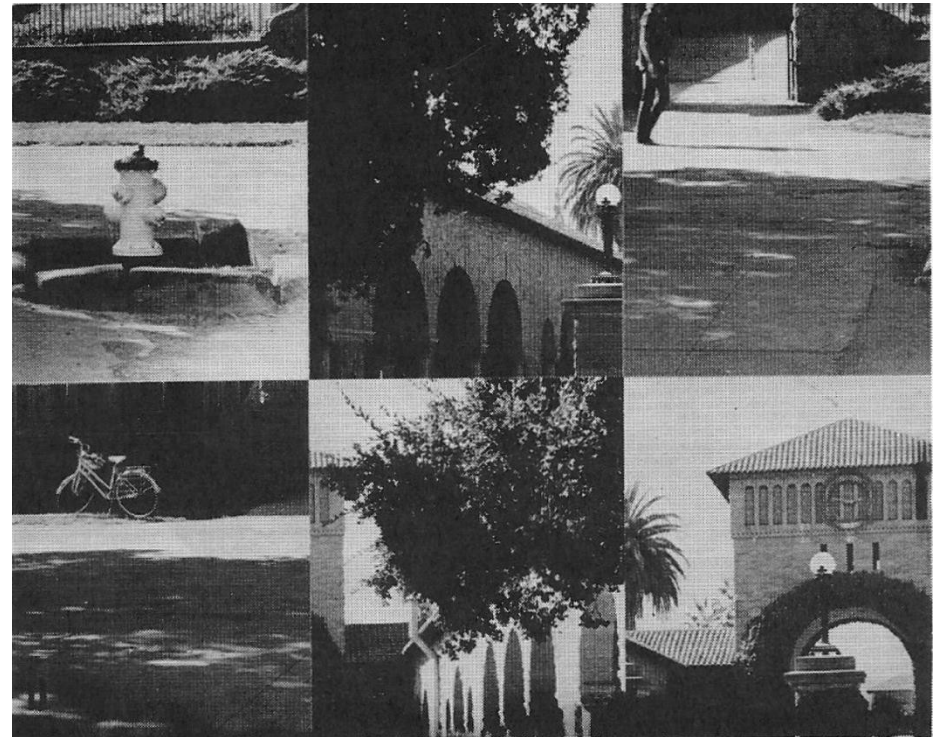
John Canny, 1986
David Lowe, 1987

In the meantime...seminal work in
cognitive and neuroscience



Perceiving Real-World Scenes

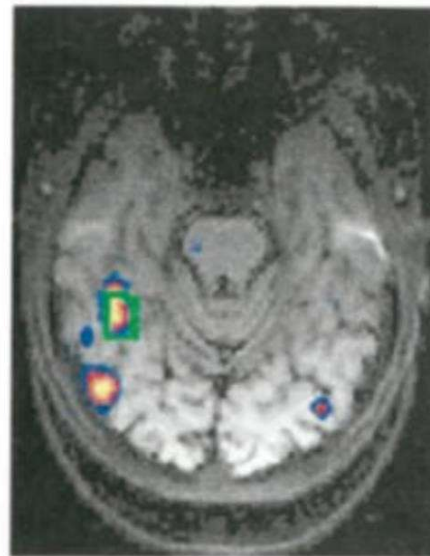
Irving Biederman



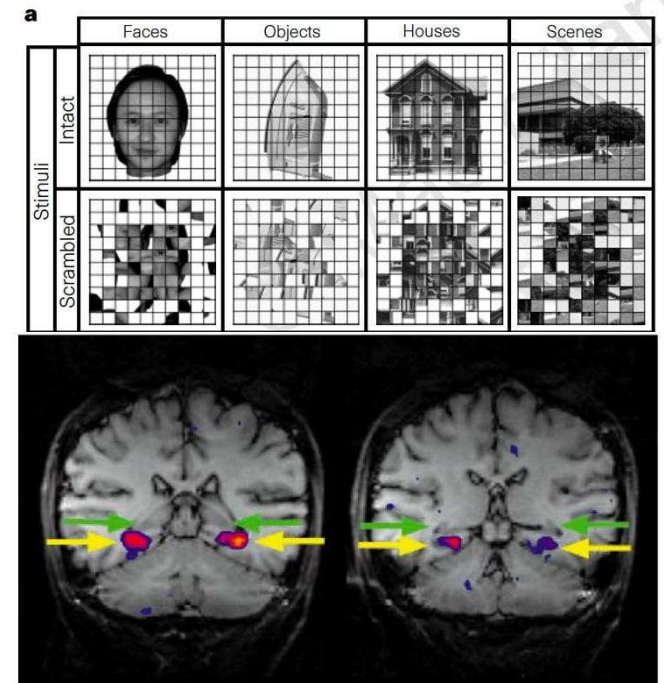
I. Biederman, *Science*, 1972

Neural correlates of object & scene recognition

Faces > Houses

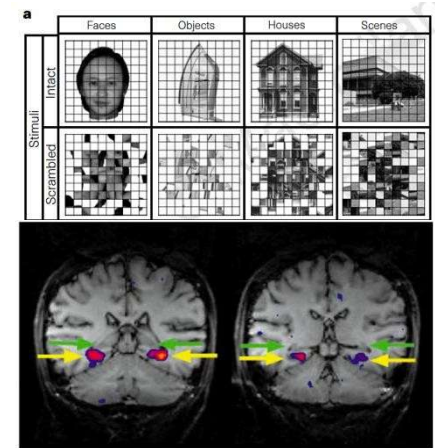


Kanwisher et al. J. Neuro. 1997

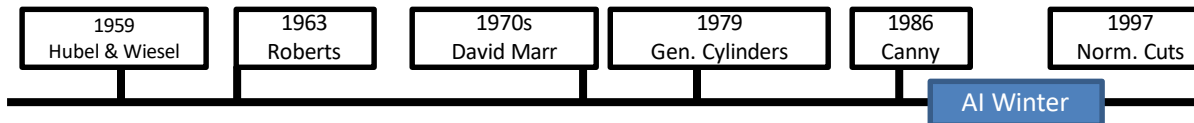


Epstein & Kanwisher, Nature, 1998

Visual recognition is a fundamental task for visual intelligence



Recognition via Grouping (1990s)



Normalized Cuts, Shi and Malik, 1997

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Right Image is CC-BY 2.0; changes made

Slide inspiration: Justin Johnson

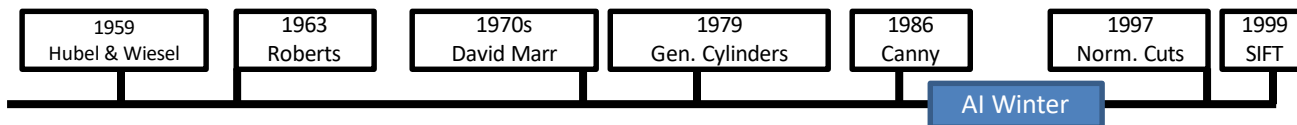
Recognition via Matching (2000s)



[Image](#) is public domain



[Image](#) is public domain



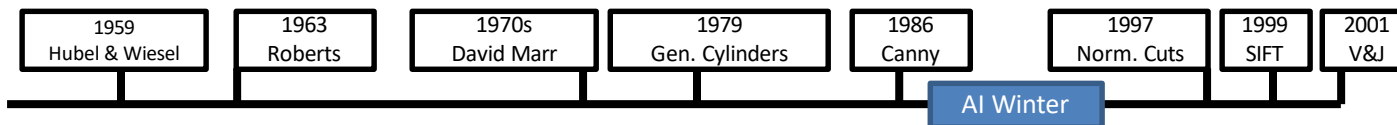
SIFT, David
Lowe, 1999

Slide inspiration: Justin Johnson

Face Detection

Viola and Jones, 2001

One of the first successful applications of machine learning to vision

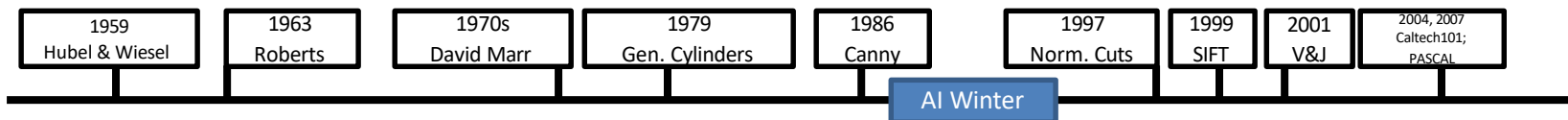


Slide inspiration: Justin Johnson

Caltech 101 images



PASCAL Visual Object Challenge



Slide inspiration: Justin Johnson

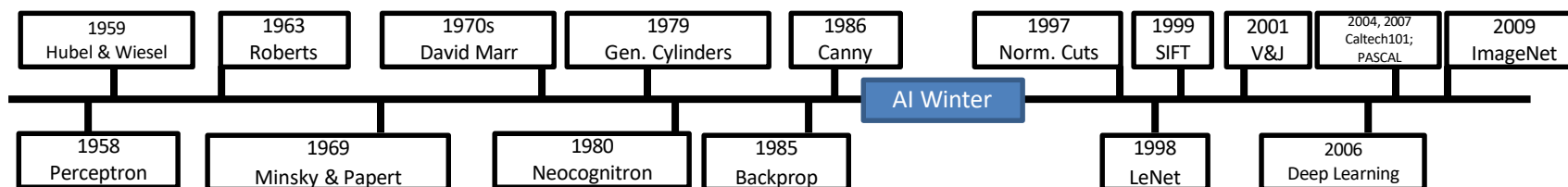
IMAGENET Large Scale Visual Recognition Challenge

The Image Classification Challenge:
1,000 object classes
1,431,167 images

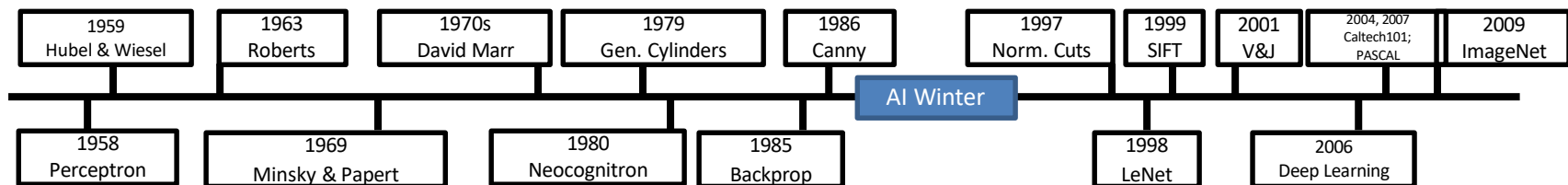
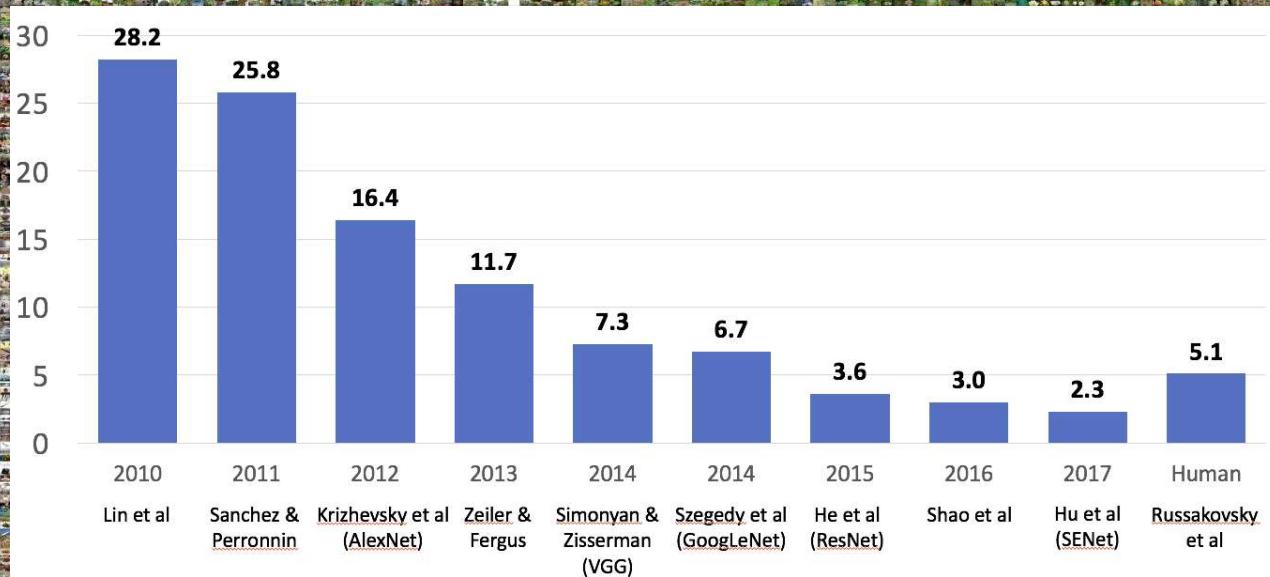


Output:
Scale
T-shirt
Steel drum
Drumstick
Mud turtle

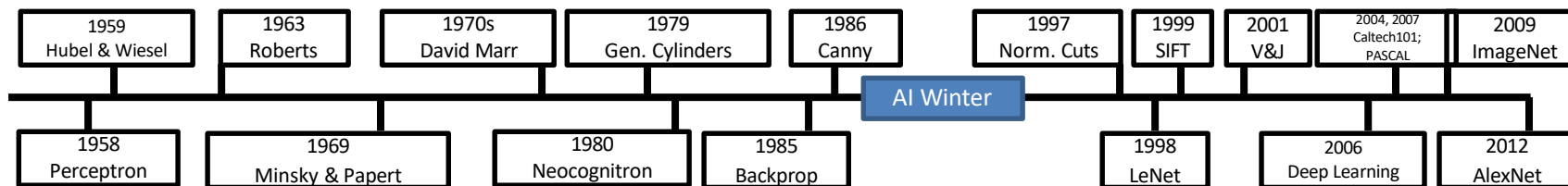
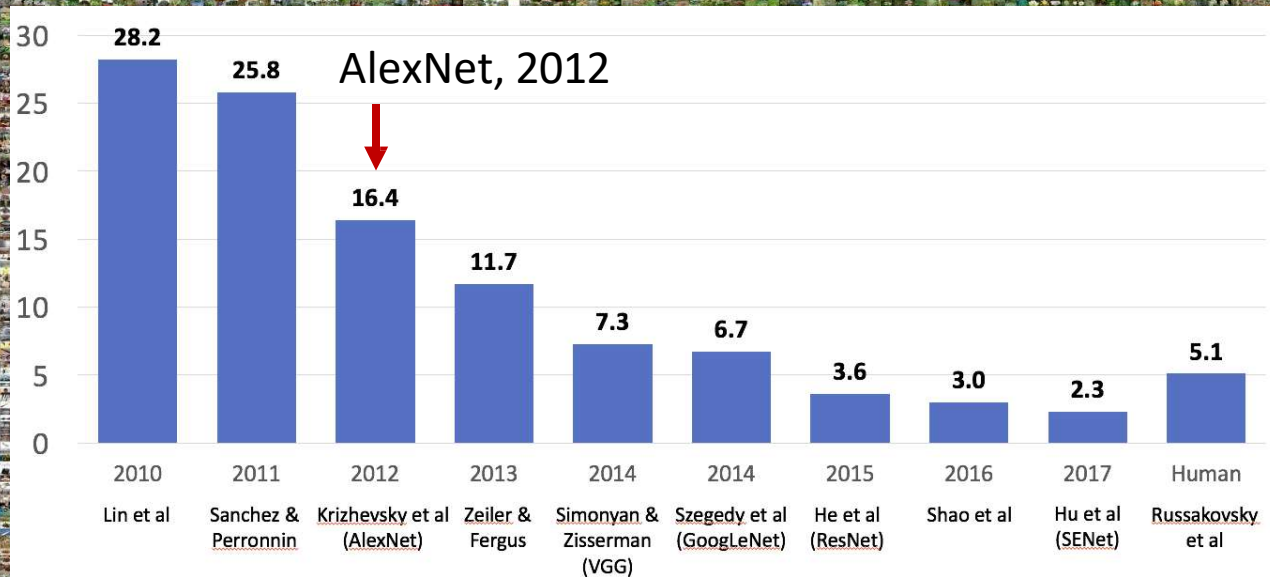
Deng et al, 2009
Russakovsky et al. IJCV 2015



IMAGENET Large Scale Visual Recognition Challenge



IMAGENET Large Scale Visual Recognition Challenge



2012 to Present: Deep Learning is Everywhere

Image Classification

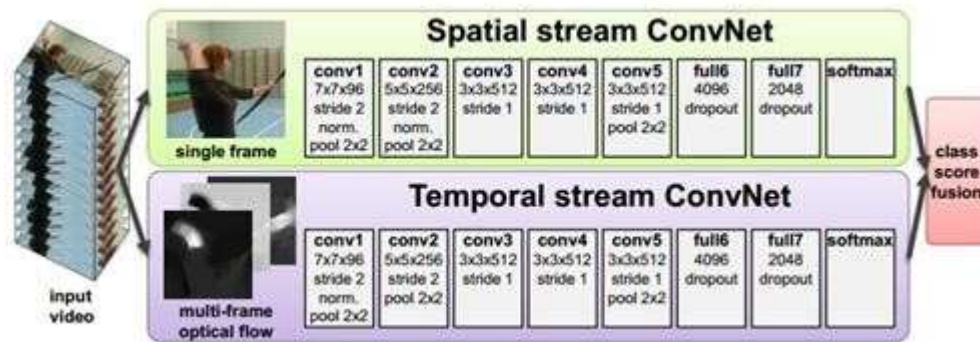
Image Retrieval



Figures copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

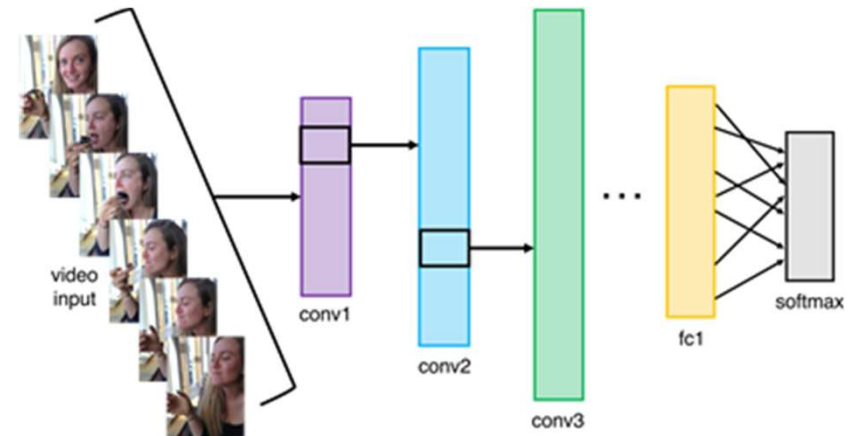
2012 to Present: Deep Learning is Everywhere

Video Classification



Simonyan et al, 2014

Activity Recognition



2012 to Present: Deep Learning is Everywhere

Pose Recognition (Toshev and Szegedy, 2014)

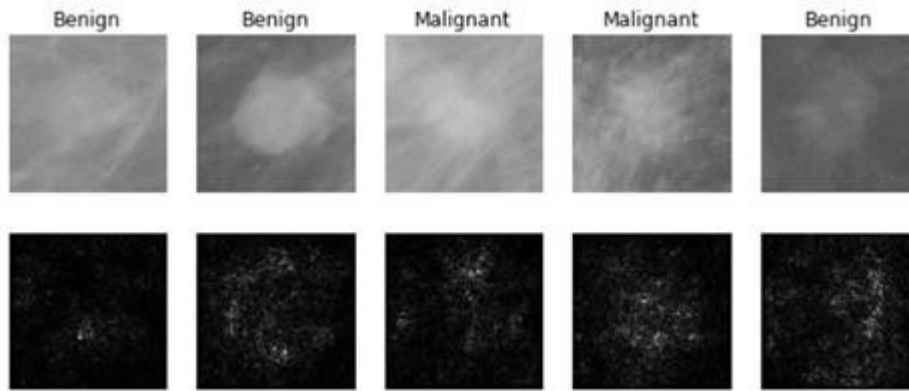


Playing Atari games (Guo et al, 2014)



2012 to Present: Deep Learning is Everywhere

Medical Imaging



Levy et al, 2016 Figure reproduced with permission

Galaxy Classification



Dieleman et al, 2014

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Whale recognition



[Kaggle Challenge](#)

This image by Christin Khan is in the public domain and originally came from the U.S. NOAA.

2012 to Present: Deep Learning is Everywhere



A white teddy bear sitting in the grass



A man in a baseball uniform throwing a ball



A woman is holding a cat in her hand

Image Captioning

Vinyals et al, 2015

Karpathy and Fei-Fei, 2015



A man riding a wave on top of a surfboard



A cat sitting on a suitcase on the floor

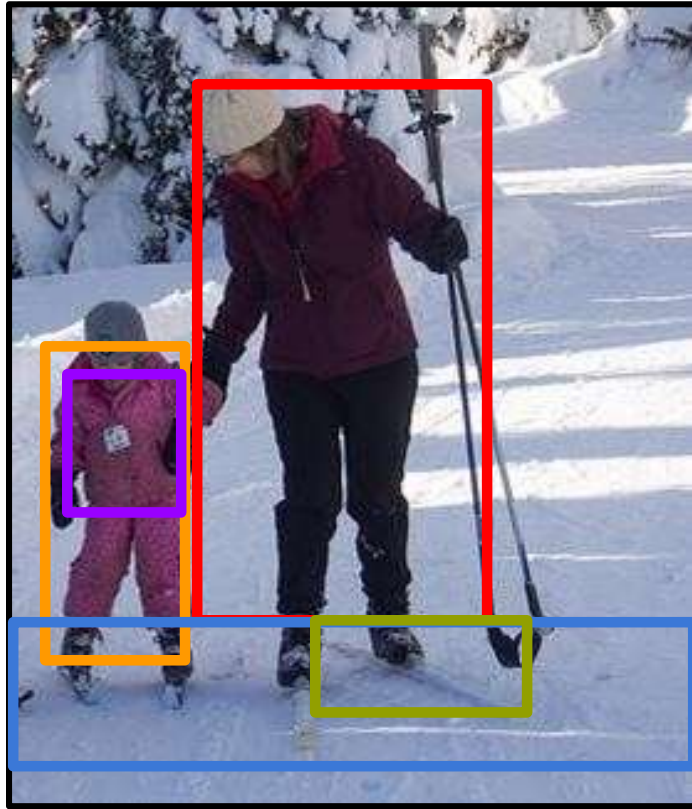


A woman standing on a beach holding a surfboard

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<https://pixabay.com/en/luggage-antique-cat-1643010/>
<https://pixabay.com/en/teddy-plush-bears-cute-teddy-bear-1623436/>
<https://pixabay.com/en/surf-wave-surfboard-sport-1668716/>
<https://pixabay.com/en/woman-female-model-porcelain-adult-983967/>
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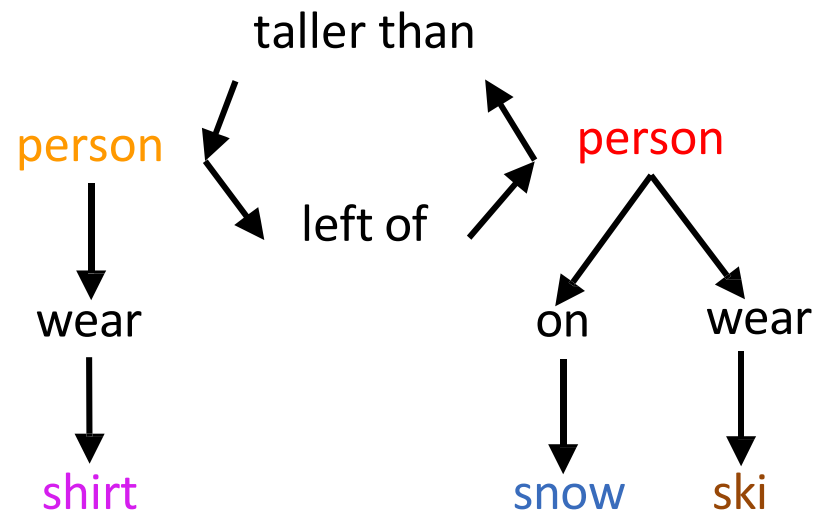
Captions generated by Justin Johnson using: [NeuralTalk2](#)

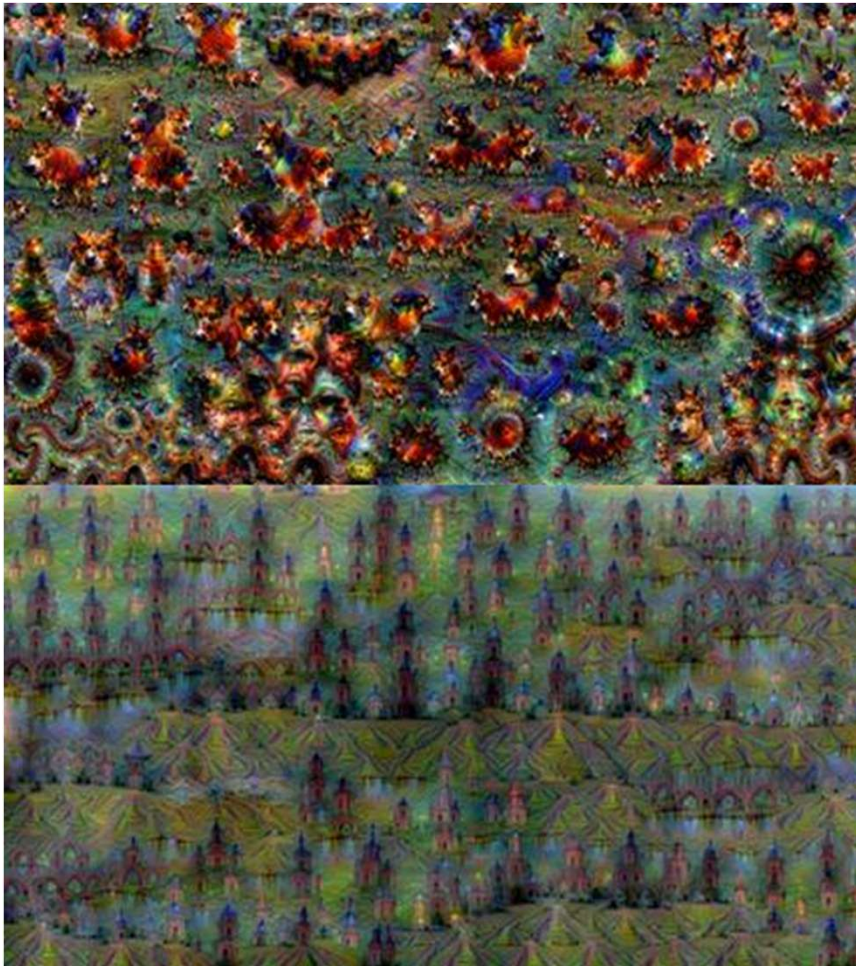
2012 to Present: Deep Learning is Everywhere



Results:

spatial, comparative, asymmetrical, verb,
prepositional





Original image: CC0 public domain
 Starry Night and Tree Roots by Van Gogh are in the public domain
 Bokeh image is in the public domain
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Mordvinsev et al, 2015
 Gatys et al, 2016

Figures copyright Justin Johnson, 2015. Reproduced with permission. Generated using the Inceptionism approach from a [blog post](#) by Google Research.

2012 to Present: Deep Learning is Everywhere

TEXT PROMPT

an armchair in the shape of an avocado. an armchair imitating an avocado.

AI-GENERATED IMAGES



2012 to Present: Deep Learning is Everywhere

TEXT PROMPT

an armchair in the shape of a peach. an armchair imitating a peach.

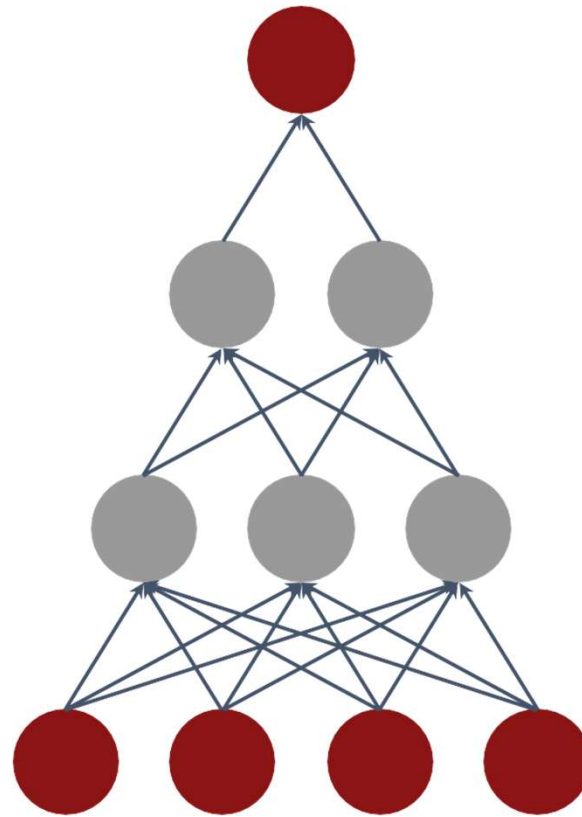
AI-GENERATED IMAGES





Computation

April 4, 2025



Algorithms



Data

AI's Explosive Growth & Impact

Attendance at large conferences (1984-2019)
Source: Conference provided data.

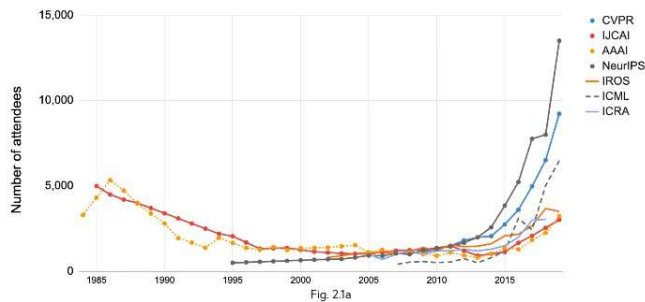
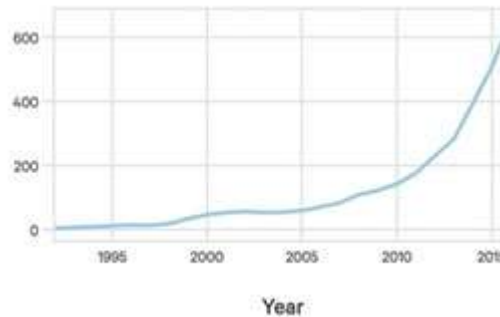


Fig. 2.1a

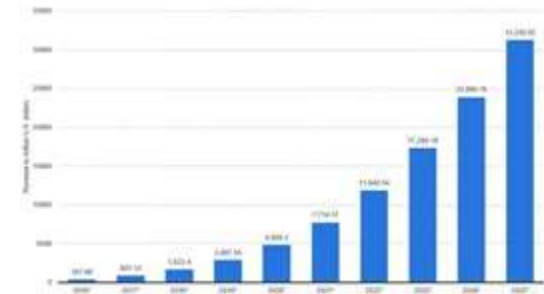
**Number of attendance
At AI conferences**

Source: The Gradient



**Startups Developing AI
Systems**

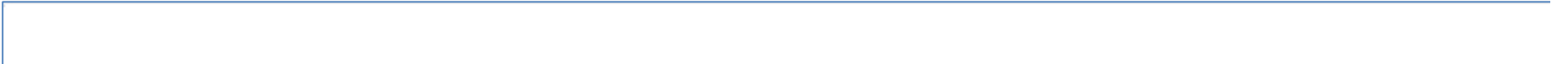
Source: Crunchbase, VentureSource, Sand
Hill Econometrics



**Enterprise Application AI
Revenue**

Source: Statista

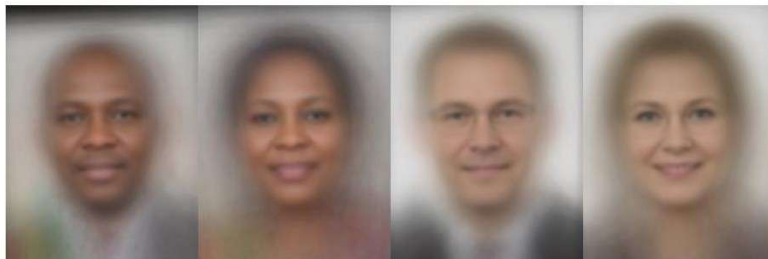
Despite the successes, computer vision still has a long way to go



Computer Vision Can Cause Harm

Harmful Stereotypes

Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
 Microsoft	94.0% 	79.2% 	100% 	98.3% 	20.8%
 FACE++	99.3% 	65.5% 	99.2% 	94.0% 	33.8%
 IBM	88.0% 	65.3% 	99.7% 	92.9% 	34.4%

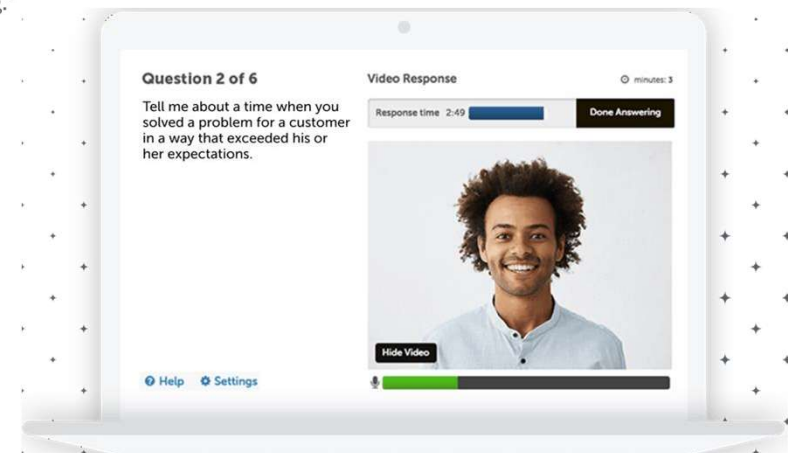


Affect people's lives

Technology

A face-scanning algorithm increasingly decides whether you deserve the job

HireVue claims it uses artificial intelligence to decide who's best for a job. Outside experts call it 'profoundly disturbing.'



Source: <https://www.washingtonpost.com/technology/2019/10/22/ai-hiring-face-scanning-algorithm-increasingly-decides-whether-you-deserve-job/>
<https://www.hirevue.com/platform/online-video-interviewing-software>

Example Credit: Timnit Gebru

Computer Vision Can Save Lives

How to take care of seniors while keeping them safe?

Early Symptom Detection of COVID-19

Monitor Patients with Mild Symptoms

Manage Chronic Conditions

Versatile

Scalable

Mobility

Infection

Sleep

Diet

Low-cost

Burden-free



And there is a lot we don't know how to do



https://fedandfit.com/wp-content/uploads/2020/06/summer-activities-for-kids_optimized-scaled.jpeg



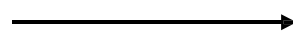
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Image Classification: A core task in Computer Vision



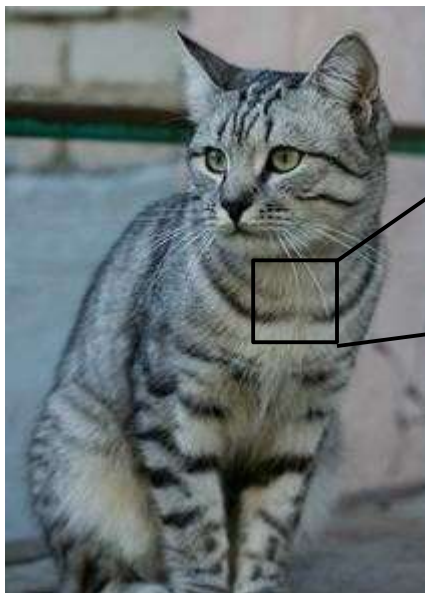
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(assume given a set of possible labels)
{dog, cat, truck, plane, ...}



cat

The Problem: Semantic Gap



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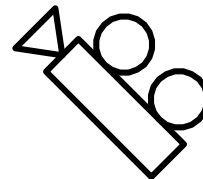
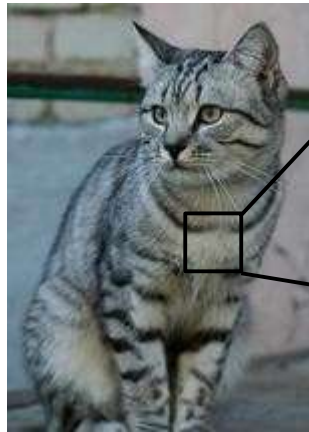
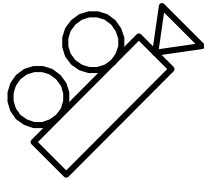
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[[105 112 108 111 104 99 106 99 96 103 112 119 104 97 93 87]
 [ 91 98 102 106 104 79 98 103 99 105 123 136 110 105 94 85]
 [ 76 85 98 105 128 105 87 96 95 99 115 112 106 103 99 85]
 [ 99 81 81 93 120 131 127 100 95 98 102 99 96 93 101 94]
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 [ 87 65 71 87 106 95 69 45 76 130 126 107 92 94 105 111]
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 [128 112 96 117 150 144 120 115 104 107 102 93 87 81 72 79]
 [123 107 96 86 83 112 153 149 123 109 104 75 88 107 112 99]
 [122 121 102 88 82 86 94 117 145 140 153 102 58 78 92 107]
 [122 164 148 103 71 56 78 83 93 103 119 139 102 61 69 84]]
```

What the computer sees

An image is a tensor of integers
between $[0, 255]$:

e.g. $800 \times 600 \times 3$
(3 channels RGB)

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All pixels change when the camera moves!

Challenges: Illumination



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Challenges: Background Clutter



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Challenges: Occlusion



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Challenges: Deformation



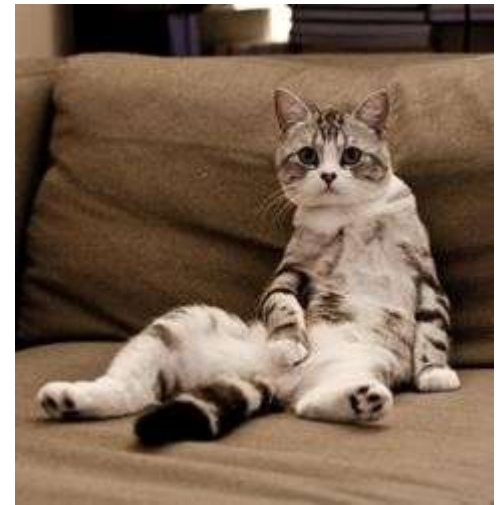
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Challenges: Intraclass variation



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Challenges: Context



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Challenges: Context

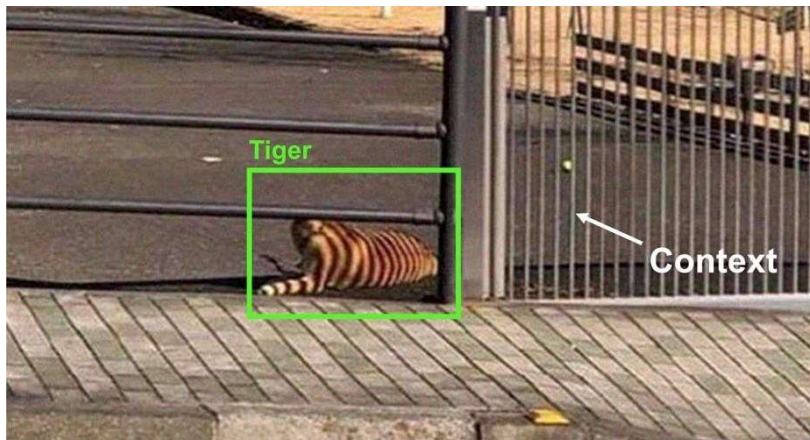


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Challenges: Context

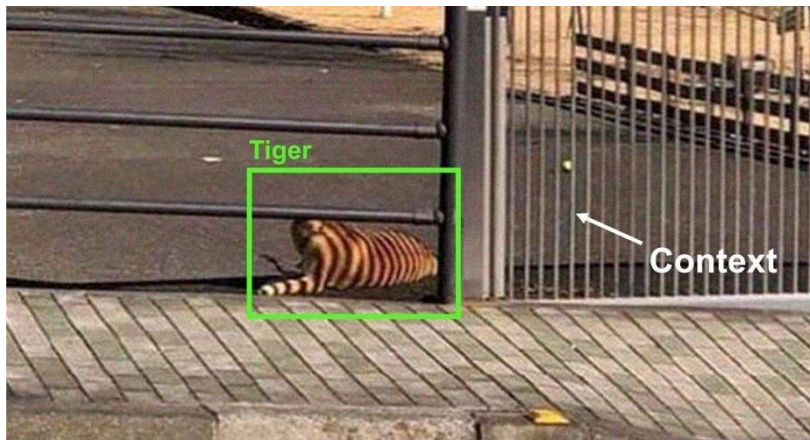


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Modern computer vision algorithms



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